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ECE F266 — Study Report

Optical Inter-Satellite Link Channel Simulation & Rate Analysis

Kramers–Kronig Self-Coherent Receiver with Transmitter-Side Carrier

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Chapter 1

Introduction

1.1 Motivation

The deployment of large low-Earth orbit (LEO) satellite constellations by commercial operators has renewed interest in high-capacity optical inter-satellite links (OISLs). Unlike radio-frequency cross-links, free-space optical (FSO) links offer licence-free operation, narrow beam divergence and data rates on the order of tens of gigabits per second [3]. These properties make FSO the preferred backhaul technology for mega-constellations that must relay traffic between orbital planes with minimal latency.

Conventional coherent fibre-optic receivers recover the full complex optical field, but they require a local oscillator (LO) laser that must be frequency-locked to the transmitter, track Doppler shifts, and match the polarisation state of the incoming signal. For a satellite terminal operating in a 550 km LEO shell, the carrier Doppler shift between two spacecraft in different planes can reach ± 9.9 GHz; the LO must follow this shift in real time while maintaining phase coherence at symbol rates of 10 GBaud or higher. The hardware burden—a tunable LO laser, a polarisation-diversity front end, and the associated control loops—adds mass, power consumption and cost to every spacecraft.

The Kramers–Kronig (KK) self-coherent receiver, first proposed by Mecozzi, Antonelli and Shtaif [1], offers an alternative. By transmitting a continuous-wave (CW) carrier alongside the single-sideband (SSB) data signal and requiring only a single photodiode, it recovers the optical field entirely in digital signal processing (DSP). When the carrier is placed at the *transmitter*, it co-propagates with the data through the same laser, the same spatial mode and the same polarisation. At the receiver, square-law detection causes the common laser phase noise, Doppler shift and polarisation rotation to cancel automatically. The resulting receiver is a photodiode (optionally preceded by an erbium-doped fibre amplifier, EDFA) followed by an analogue-to-digital converter and a DSP chain: the simplest coherent-capable architecture possible for an OISL terminal.

1.2 Scope and Objectives

This study report develops an end-to-end analytical model of a KK-OISL link with a transmitter-side carrier, derives the achievable data rates under realistic pointing-error fading, and validates the model through Monte Carlo simulation. The specific objectives are:

1. Formulate the optical field model for a KK-OISL transmitter and receiver, including the carrier–signal power split (CSPR), single-sideband modulation and the minimum-phase

condition.

2. Derive the electrical signal-to-noise ratio (SNR) at the receiver output, accounting for five physical noise sources (shot noise, thermal noise, laser relative intensity noise (RIN), signal/carrier–ASE beat noise and ASE×ASE beat noise) and the noise enhancement of the KK algorithm.

3. Integrate the SNR model over the pointing-error fading distribution to obtain ergodic and outage data rates for BPSK, OOK and QPSK modulation formats.

4. Validate the analytical model against two levels of Monte Carlo simulation: a channel-level simulation that samples the pointing fading, and a waveform-level simulation that generates pulse-shaped symbols, adds all physical noise sources, runs the full KK DSP chain, and measures the output SNR and bit-error rate.

5. Investigate the effect of carrier/LO placement by simulating four configurations and showing that KK phase retrieval fails without a carrier at the photodiode input.

6. Package the model as a self-contained MATLAB function suitable for co-simulation with commercial optical link-design tools such as OptiSystem.

1.3 Contributions

The main contributions of this work are:

- A waveform-validated noise model for the transmitter-carrier KK receiver under pointing-error fading. Each noise term is checked separately against an end-to-end waveform simulation; the model holds within 0.15 dB for CSPR $\kappa \geq 12$ dB. A closed-form Gaussian bound on the KK reconstruction penalty is shown to be 35–50 dB too pessimistic, so the optimum CSPR is taken from the waveform simulation instead: about 7 dB on-axis and 9 dB in deep fades.

- A constellation-level comparison of four carrier/LO configurations (transmitter carrier, receiver-side optical LO, no carrier, and a carrier added digitally after detection) with a clear physical explanation of why the last two fail: square-law detection destroys optical phase irreversibly, and the KK Hilbert-transform relation holds only when the field is minimum-phase.

- A parametric study of detector aperture radius (0.04–0.10 m) and ITU-T C-band wavelength (1530 and 1565 nm), showing that the default detector radius of 0.07 m reaches about 94 % of the rate obtained with the largest radius in the sweep (0.10 m), and that the rate varies by about 2 % across the C-band.

- A simulation package for MATLAB and GNU Octave that needs no toolboxes, with regression checks for both the analytic model and the waveform simulation, and a callable analytic model for OptiSystem co-simulation.

1.4 Report Organisation

The remainder of this report is organised as follows. Chapter 2 reviews the relevant background: free-space optical channel models, pointing-error statistics and the Kramers–Kronig receiver. Chapter 3 presents the mathematical derivation of the KK-OISL model, the simulation results, and the rationale for the transmitter-side carrier. Chapter 4 summarises the findings and identifies directions for future work.

Chapter 2

Background and Literature Review

This chapter reviews the three bodies of work on which the present model is built: free-space optical link budgets and pointing-error statistics, self-coherent receivers based on the Kramers–Kronig relation, and the particular challenges of applying coherent detection to inter-satellite links.

2.1 Free-Space Optical Channel

2.1.1 Link Geometry and Deterministic Loss

A Gaussian laser beam of waist w_0 and wavelength $\lambda = c/f$ diverges in the far field with a half-angle $\theta = \lambda/(\pi w_0)$. At a propagation distance z the beam radius is [3]:

$$w_z = z\tan\Big(\frac{\lambda}{\pi w_0}\Big) \approx \frac{z\lambda}{\pi w_0} \tag{2.1}$$

A circular receive aperture of radius w_d centred on the beam axis collects a fraction

$$A_0 = \frac{2w_d^2}{w_z^2} \tag{2.2}$$

of the total transmitted power, where the approximation assumes a Gaussian intensity profile at the receiver. The maximum received power in the absence of pointing error is therefore $P_{rx,\max} = h_{PL}A_0P_T$, where h_{PL} accounts for deterministic path losses (telescope throughput, dichroic filters and fibre coupling).

2.1.2 Pointing-Error Fading

The satellite pointing jitter, due to attitude-control residuals, structural vibration and thermal distortion, is modelled as a random radial displacement r of the beam centre from the receiver aperture. For two-axis jitter with equal variance σ_s^2, the displacement is Rayleigh distributed: $f_R(r) = (r/\sigma_s^2)\exp(-r^2/2\sigma_s^2)$ [2].

The collected power fraction as a function of radial displacement is approximately $h_{PE}(r) = A_0\exp(-2r^2/w_z^2)$. Defining the normalised fading variable $y = h_{PE}(r) \in (0, A_0]$, its probability density function and cumulative distribution function follow a power law [2]:

$$f_Y(y) = \frac{\alpha}{A_0^\alpha}y^{\alpha-1}, \qquad F_Y(y) = \Big(\frac{y}{A_0}\Big)^\alpha, \qquad 0 < y \le A_0 \tag{2.3}$$

where the shape parameter is

$$\alpha = \frac{w_z^2}{4\sigma_s^2} \tag{2.4}$$

A small α (large jitter relative to the beam) produces heavy-tailed fading with frequent deep fades. With the jitter model and parameters used in this report, $\alpha = 0.20$ at 1000 km (a stress case in which the jitter exceeds the beam radius); $\alpha \approx 3$ would require a base jitter $\sigma_{s0} \approx 0.5$ m.

The jitter standard deviation may itself depend on range. Shang *et al.* [2] use the model $\sigma_s(z) = \sigma_{s0}\exp(k_0 z/d_0)$, where σ_{s0}, k_0 and d_0 are platform-dependent constants. This captures the physical observation that fine pointing becomes harder as the angular subtense of the beacon shrinks with distance.

2.1.3 Data Rate over a Fading Channel

Given a receiver SNR model $g(y)$ that depends on the instantaneous fading state y, the ergodic achievable rate is the average of the mutual information over the fading distribution:

$$R = B\int_0^{A_0} m\big(g(y)\big)\, f_Y(y)\, dy \tag{2.5}$$

where B is the symbol rate and $m(g)$ is the spectral efficiency at SNR g (in bits per symbol per hertz). For a memoryless AWGN channel with hard-decision decoding, the spectral efficiency is bounded by $m(g) = k[1 - H_2(\mathrm{BER}(g))]$, where k is the number of bits per symbol and H_2 is the binary entropy function.

The ε-outage rate, defined as the rate that can be maintained for all but a fraction ε of the fading realisations, follows from inverting the CDF:

$$R_\varepsilon = B \cdot m\big(g(A_0\,\varepsilon^{1/\alpha})\big) \tag{2.6}$$

This closed-form expression is a direct consequence of the power-law CDF in eq. (2.3).

2.2 Self-Coherent Detection and the Kramers–Kronig Receiver

2.2.1 Coherent vs Direct Detection

A conventional coherent receiver mixes the incoming signal with a local oscillator (LO) laser using a 90° optical hybrid and balanced photodiodes. This recovers the full complex optical field (amplitude and phase) and enables the highest spectral efficiencies. However, it requires the LO to be frequency-locked to the transmitter, demands polarisation alignment (or a diversity receiver) and needs carrier-phase recovery in the DSP.

Direct detection with a single photodiode is simpler but measures only the intensity $|E|^2$, discarding the phase. For intensity-modulated formats such as on–off keying (OOK), this suffices; for phase-carrying formats such as QPSK, the phase information is lost.

2.2.2 The KK Principle

The Kramers–Kronig receiver bridges these two extremes [1]. It uses a single photodiode but recovers the optical phase from the detected intensity by exploiting a property of minimum-phase signals: the phase and the logarithm of the amplitude are related by the Hilbert transform.

A signal $E(t)$ is minimum-phase if its complex analytic extension has no zeros in the upper half of the complex plane. A sufficient condition for a carrier-plus-signal field $E(t) = \sqrt{P_c} + \sqrt{P_s}\, s(t)\, e^{i2\pi f_{sh} t}$ is that the carrier amplitude exceeds the peak of the modulated signal:

$$\sqrt{P_c} > \sqrt{P_s}\, |s(t)|_{\max} \quad \Longleftrightarrow \quad \kappa > \mathrm{PAPR}_s \tag{2.7}$$

where $\kappa = P_c/P_s$ is the carrier-to-signal power ratio and PAPR_s is the peak-to-average power ratio of the baseband signal $s(t)$. When this condition is satisfied, the optical phase can be reconstructed as:

$$\phi_E(t) = \mathscr{H}\Big\{ \tfrac{1}{2} \ln I(t) \Big\} \tag{2.8}$$

where $I(t)$ is the photocurrent (in power units) and $\mathscr{H}\{\cdot\}$ denotes the Hilbert transform. The full complex field is then $\hat{E}(t) = \sqrt{I(t)}\, e^{i\phi_E(t)}$.

The KK receiver was first demonstrated in a fibre-optic link at 200 Gb/s using 16-QAM [1]. It has since been extended to higher-order modulation formats, wavelength-division multiplexed systems, and longer reaches.

2.2.3 Carrier Placement: Transmitter vs Receiver

In fibre-optic applications the CW carrier can be generated at either end of the link:

- **Transmitter-side carrier.** The carrier is generated from the same laser as the data and transmitted in the same spatial mode. The CSPR is fixed at the transmitter and cannot be adjusted independently. The advantage is that the carrier and signal experience identical channel impairments; any common-mode effect (phase noise, chromatic dispersion) cancels in detection.
- **Receiver-side LO.** A local CW laser is combined with the incoming signal before the photodiode. The LO power can be set freely, and the CSPR can be made arbitrarily large. However, the LO and the transmitter laser are independent, so their phase noise, frequency offset and polarisation mismatch must be tracked and compensated.

In a fibre link the distinction is largely one of convenience: the channel is stable, the frequency offset is small, and polarisation can be tracked at moderate complexity. In a free-space inter-satellite link, however, the distinction is fundamental, as discussed in the next section.

2.3 Challenges of Coherent Detection in OISL

Several properties of the OISL channel make receiver-side LO architectures significantly more complex than their fibre-optic counterparts:

Doppler shift. Two LEO spacecraft in different orbital planes can have a relative radial velocity of up to ± 15 km/s. At $\lambda = 1530$ nm this corresponds to a carrier Doppler shift of approximately ± 9.9 GHz. A receiver-side LO must track this shift in real time, requiring either a widely tunable laser (acousto-optic or electro-optic frequency shifter) or a very high-bandwidth digital frequency estimator.

Phase noise. Two independent lasers contribute uncorrelated phase noise to the beat signal. At the 10 GBaud symbol rates considered here, linewidths above approximately 100 kHz begin to cause significant cycle-slip rates. Space-qualified narrow-linewidth lasers are available but add to the terminal's mass and power budget.

Polarisation. The received optical field arrives in a time-varying polarisation state that depends on spacecraft attitude, telescope alignment and thermal flexure. A single-photodiode receiver with a receiver-side LO sees only the component of the signal that projects onto the LO's polarisation. Full recovery requires a polarisation-diversity front end (a polarisation beam splitter, two photodiodes, and Stokes-parameter tracking), roughly doubling the front-end hardware.

Beam wander and scintillation. Unlike the fibre channel, the free-space channel subjects the beam to pointing jitter (Section 2.1.2). The fading reduces the received signal power but does not alter the CSPR when carrier and signal co-propagate. With a receiver-side LO the CSPR varies inversely with the fading, because the LO power is fixed.

These considerations motivate the use of a *transmitter-side* carrier for the KK-OISL link. Because the carrier and signal share the same laser and propagation path, the Doppler shift, phase noise and polarisation rotation cancel in square-law detection (Chapter 3, eq. 3.5). The receiver is reduced to a photodiode, an optional EDFA and digital back-end—the minimal hardware configuration for a coherent-capable OISL terminal.

2.4 Related Work

The analytical framework for OISL channel modelling and rate analysis follows Shang, Zhang and Wong [2], who derived ergodic and outage rates for a coherent OISL link under power-law pointing-error fading. The present work extends their model by replacing the conventional coherent receiver with a KK self-coherent receiver, adding the CSPR power split, the KK noise enhancement factor and the reconstruction penalty.

The KK receiver was introduced by Mecozzi, Antonelli and Shtaif [1] for fibre-optic links and has been studied extensively in that context [4]. Its application to free-space links is more recent. Naghshvarianjahromi, Kumar and Deen [5] analysed a KK transceiver with a transmitter-side carrier for GEO–GEO inter-satellite and ground–satellite links under atmospheric turbulence and compared it with coherent detection. Their analysis does not include satellite pointing-jitter fading or ergodic and outage rates. The present work addresses that gap for LEO inter-satellite links: it combines the pointing-error fading model of Shang *et al.* with a physical noise budget of the transmitter-carrier KK receiver, validates the model against an end-to-end waveform simulation, and determines the optimum CSPR under fading. Wen *et al.* [6] showed that KK receivers are more sensitive than coherent receivers to front-end nonlinearity and memory and proposed memory-polynomial pre-compensation (MP-KK); their measured optimum CSPR of 7–8 dB agrees with the waveform results of Chapter 3.

Chapter 3

Integration of Kramers–Kronig Architecture in Optical Inter-Satellite Link Model

3.1 Mathematical Derivation of KK-OISL Model

3.1.1 Optical Field Representation and Minimum-Phase Condition

The Kramers–Kronig (KK) receiver needs a strong continuous-wave (CW) tone at the edge of the signal spectrum [1]. In this model the tone (carrier) is generated at the *transmitter* from the same laser and IQ modulator as the data, and travels with the data over the link. A baseband data signal $s(t)$ of symbol rate B, normalised to unit average power, is single-sideband (SSB) shifted to the right of the carrier:

$$E_{Tx}(t) = \left[\sqrt{P_c} + \sqrt{P_s}\, s(t)\, e^{i2\pi f_{sh} t}\right] e^{i\theta_L(t)}, \qquad f_{sh} = B_o/2 \tag{3.1}$$

where $B_o = (1+\beta)B$ is the occupied bandwidth for a pulse roll-off β, $\theta_L(t)$ is the laser phase noise, P_c is the carrier power and P_s is the signal power. The SSB term occupies $[0, B_o]$, so the carrier lies at the lower edge of the signal spectrum.

The cross term $2\sqrt{P_c P_s}\,\mathrm{Re}\{s\, e^{i2\pi f_{sh} t}\}$ averages to zero, so the total transmit power is $P_T = P_c + P_s$. With the carrier-to-signal power ratio (CSPR) $\kappa = P_c/P_s$:

$$P_c = \frac{\kappa}{1+\kappa} P_T, \qquad P_s = \frac{1}{1+\kappa} P_T \tag{3.2}$$

Only the fraction $1/(1+\kappa)$ of the transmit power carries data. This is the physical CSPR penalty.

At the receiver aperture both components are attenuated by the same total channel power gain $h = h_{PL} h_{PE}$:

$$E_{Rx}(t) = \sqrt{h}\left[\sqrt{P_c} + \sqrt{P_s}\, s(t)\, e^{i2\pi f_{sh} t}\right] e^{i\theta_L(t) + i\phi_D(t)} \tag{3.3}$$

where $\phi_D(t)$ is the Doppler phase. Carrier and data come from the same laser and travel the same path in the same polarisation and spatial mode. The laser phase noise, the Doppler shift, any polarisation rotation and any wavefront tilt are therefore common to both and cancel in square-law detection (eq. 3.5). As a result:

- No local-oscillator frequency locking is needed. For a 550 km LEO shell the carrier Doppler can reach ±9.9 GHz. Carrier and data are shifted almost equally; the data sideband, which lies $f_{sh} = 5.25$ GHz above the carrier, receives an extra shift of $f_{sh}v/c \approx 263$ kHz at $v = 15$ km/s, and the symbol clock is scaled by $v/c = 50$ ppm. Without correction these residuals destroy the link (SNR ≈ 2 dB in the waveform simulation). A pilot-based frequency estimate, a clock correction derived from the same estimate ($\varepsilon = \Delta f / f_{sh}$) and blind phase tracking remove them, with a penalty of at most 0.4 dB down to the HD-FEC threshold.
- No polarisation control is needed.
- The CSPR at the receiver equals κ for every pointing state.

For exact phase retrieval from intensity, $E_{Rx}(t)$ must be minimum phase. A sufficient condition is that the time trajectory of the field does not encircle the origin:

$$\sqrt{P_c} > \sqrt{P_s}\,|s(t)|_{\max} \quad \Longleftrightarrow \quad \kappa > \mathrm{PAPR}_s = \frac{\max|s(t)|^2}{\langle |s(t)|^2 \rangle} \tag{3.4}$$

Proof. Put $u = \sqrt{P_s/P_c}\, s\, e^{i2\pi f_{sh}t}$, with $|u| < 1$. Then $\ln(1+u) = \sum_n (-1)^{n+1} u^n/n$ contains only positive frequencies, so its imaginary part (the phase) is the Hilbert transform of its real part, $\ln|1+u|$ [1]. □

For root-raised-cosine (RRC) pulses with $\beta = 0.05$, the PAPR, defined as the 10^{-4} point of the instantaneous-power CCDF (2^{17} symbols), is about 7.5 dB (BPSK), 7.4 dB (OOK) and 6.4 dB (QPSK). The largest peak of a single waveform is not used, because it depends on the random sequence. $\kappa = 12$ dB is used throughout; the noise model of Section 3.1.3 is verified for $\kappa \geq 12$ dB.

3.1.2 Square-Law Detection and KK Phase Reconstruction

A photodiode with responsivity η performs square-law detection to produce the photocurrent $I(t)$:

$$I(t) = \underbrace{\eta h P_c}_{\text{Carrier DC Current}} + \underbrace{\eta h P_s |s(t)|^2}_{\text{Signal-Signal Beat Current}} + \underbrace{2\eta h \sqrt{P_c P_s}\,\mathrm{Re}\{s(t) e^{i2\pi f_{sh}t}\}}_{\text{Cross-Beat Signal Current}} + n(t) \tag{3.5}$$

The common phases θ_L and ϕ_D have disappeared. The photocurrent is converted to power units by dividing by the responsivity only. No noise is subtracted, and no division by the unknown channel gain is needed:

$$I_n(t) = \frac{I(t)}{\eta} \tag{3.6}$$

Under the minimum-phase condition, the optical phase $\phi_E(t)$ and the field envelope $\sqrt{I_n(t)}$ are uniquely related by the KK relation, through the Hilbert transform $\mathscr{H}\{\cdot\}$:

$$\phi_E(t) = \mathscr{H}\left\{\tfrac{1}{2}\ln I_n(t)\right\} = \frac{1}{2\pi}\,\mathrm{P.V.}\int_{-\infty}^{\infty} \frac{\ln I_n(t')}{t - t'}\,dt' \tag{3.7}$$

With the Fourier convention $\mathscr{F}\{x\}(f) = \int x(t) e^{-i2\pi f t} dt$ (as used by MATLAB `fft`), the phase reconstruction in the frequency domain is:

$$\tilde{\phi}_E(f) = -i \cdot \mathrm{sgn}(f) \cdot \mathscr{F}\left\{\tfrac{1}{2}\ln I_n(t)\right\} \tag{3.8}$$

The complex envelope is reconstructed as:

$$\hat{E}(t) = \sqrt{I_n(t)}\, e^{i\phi_E(t)} = \sqrt{h}\Big[\sqrt{P_c} + \sqrt{P_s}\, s(t)\, e^{i2\pi f_{sh} t}\Big] \tag{3.9}$$

The received carrier is the time average $\hat{a} = \langle \hat{E}(t) \rangle = \sqrt{hP_c}$. It is removed, and *then* the result is frequency down-shifted:

$$\hat{E}_s(t) = \big[\hat{E}(t) - \hat{a}\big] e^{-i2\pi f_{sh} t} = \sqrt{hP_s}\, s(t) \tag{3.10}$$

The signal-signal beat current is removed exactly by this reconstruction; it does not have to be negligible. For a transmitted signal $s(t) = \sum_k a_k\, p(t - kT)$, matched filtering with $p(t)$ and sampling at symbol period T gives:

$$y_k = \big(\hat{E}_s * p\big)(kT) \approx \sqrt{hP_s}\, a_k \tag{3.11}$$

$$\hat{a}_k = \arg\min_c |y_k - \sqrt{hP_s}\, c|, \qquad \text{e.g. BPSK: } \hat{a}_k = \mathrm{sgn}[\mathrm{Re}(y_k)] \tag{3.12}$$

No carrier-phase recovery is needed, because the transmitted carrier is the phase reference.

3.1.3 Physical Receiver Noise Budget and Electrical SNR Model

The receiver is a PIN photodiode with thermal noise. An optional optical pre-amplifier (EDFA with fixed gain G, noise figure F and an optical band-pass filter of width B_f) can be placed before it; without the pre-amplifier, $G = 1$. The carrier and signal powers at the photodiode are $P_{c,d} = G\frac{\kappa}{1+\kappa} h_{PL}\, y\, P_T$ and $P_{s,d} = G\frac{1}{1+\kappa} h_{PL}\, y\, P_T$, where $y = h_{PE}$. The cross-beat electrical signal power is:

$$S(y) = 2\eta^2 P_{c,d} P_{s,d} = 2\eta^2 G^2 h_{PL}^2 P_T^2 y^2\, \frac{\kappa}{(1+\kappa)^2} \tag{3.13}$$

KK noise mapping. Linearising eqs. (3.7)–(3.9) for a small photocurrent noise w (in W) gives the field error $\delta = \frac{1}{2}\hat{E}\, \mathscr{A}\{w/|\hat{E}|^2\}$, where $\mathscr{A}\{x\} = x + i\mathscr{H}\{x\}$. For large κ this reduces to $\delta \approx \mathscr{A}\{w\}/(2\sqrt{P_{c,d}})$. A photocurrent noise of one-sided power spectral density (PSD) N_i [A^2/Hz] therefore gives a symbol SNR $g_i = S/(\zeta N_i B)$. The factor

$$\zeta = 1 + 2/\kappa$$

is the second-order KK noise enhancement; it was checked by waveform simulation to within 0.15 dB for $\kappa \geq 12$ dB (0.8 dB at $\kappa = 9$ dB and 4.7 dB at 6 dB).

ASE noise. The amplified spontaneous emission (ASE) PSD per polarisation is $S_{ASE} = h\nu(FG-1)/2$. ASE lying on both sides of the carrier beats with it into the same band, the left half appearing as an image. The ASE limit is therefore $g_{ASE} = P_{s,d}/[(1+\zeta)S_{ASE}B]$. The optical filter width $B_f = 30$ GHz accommodates the occupied bandwidth plus the Doppler range: $B_o + 2 \times 9.9\,\mathrm{GHz} \approx 30.3$ GHz.

The receiver noise power spectral densities are:

- **Shot Noise:** $N_{sh} = 2q\eta\,(P_{c,d} + P_{s,d} + 2S_{ASE}B_f)$
- **Front-End Thermal Noise:** $N_{th} = i_{th}^2$
- **Transmit-Laser Relative Intensity Noise (RIN):** $N_{rin} = \mathrm{RIN} \cdot \eta^2 (P_{c,d} + P_{s,d})^2$
- **ASE×ASE Beat (pre-amplifier, both polarisations):** $N_{aa} = 4\eta^2 S_{ASE}^2 (B_f - B/2)$

KK reconstruction penalty (heuristic, not used). The linear noise-mapping analysis above assumes the minimum-phase condition $\kappa > \mathrm{PAPR}_s$ holds at every instant. When κ approaches or falls below PAPR_s, the intensity $I_n(t)$ in eq. (3.7) reaches near-zero values; the logarithm amplifies these events into large phase errors that are not captured by the first-order expansion. A simple closed-form attempt is to add an inverse-SNR term set by the probability that a Gaussian approximation of the signal envelope violates the minimum-phase condition:

$$\frac{1}{g_{recon}} = \mathrm{PAPR}_s \cdot P_{viol}(\kappa), \qquad P_{viol} = \begin{cases} e^{-\kappa} & \text{QPSK (complex Gaussian)} \\ \mathrm{erfc}\left(\sqrt{\kappa/2}\right) & \text{BPSK, OOK (real Gaussian)} \end{cases} \tag{3.14}$$

For QPSK, $s(t)$ is treated as circularly symmetric complex Gaussian, so $P_{viol} = \Pr(|s|^2 > \kappa) = e^{-\kappa}$; for BPSK and OOK, $s(t)$ is treated as real Gaussian, $P_{viol} = \Pr(|s| > \sqrt{\kappa}) = \mathrm{erfc}(\sqrt{\kappa/2})$. The waveform simulation of Section 3.2 shows that this term is 35–50 dB too pessimistic: with all noise switched off, KK reconstruction of QPSK reaches 44.8, 78.2 and 100.9 dB SNR at $\kappa =$ 6, 9 and 12 dB, where the formula predicts 10.5, 27.7 and 62.0 dB. The Gaussian approximation counts how often the envelope exceeds the carrier but not how strongly the occasional, very short violations disturb the reconstructed field. The term is therefore *not* included in the results; the code keeps it as an option (`P.recon`) for comparison, and the optimum CSPR is determined by waveform simulation.

The SNR contribution of each noise is:

$$g_j(y) = \frac{S(y)}{\zeta N_j B} \quad (j = sh, th, rin, aa), \qquad g_{ASE}(y) = \frac{P_{s,d}}{(1+\zeta) S_{ASE} B} \tag{3.15}$$

The independent noise powers add, so the total average electrical SNR as a function of the pointing fraction y is:

$$g(y) = \left(\sum_j \frac{1}{g_j(y)} + \frac{1}{g_{recon}} \right)^{-1} \tag{3.16}$$

where the sum runs over $j \in \{sh, th, rin, ASE, aa\}$ (the last two only with the pre-amplifier). The heuristic term $1/g_{recon}$ of eq. (3.14) is omitted (see above).

Without the pre-amplifier, the thermal term scales as y^{-2}, because the carrier arrives as weak as the data and the receiver has no local-oscillator gain. With the pre-amplifier, the ASE term ($\propto y^{-1}$) dominates near the beam axis, but below $y \approx 0.02 A_0$ the thermal term dominates again (Fig. 3.4). Deep fades, where outages occur, are therefore thermal-noise limited even with the EDFA.

The CSPR enters the photocurrent-noise terms as $\kappa/(1+\kappa)^2$ and the ASE term as $1/(1+\kappa)$. Raising κ moves power from the data to the carrier and lowers the SNR; lowering it towards PAPR_s makes KK reconstruction degrade. The optimum lies in between. Because the analytic model is verified only for $\kappa \geq 12$ dB and the heuristic reconstruction term is unreliable, the optimum is found from the waveform simulation (Section 3.2).

3.1.4 Pointing Error Statistics and Fading Distribution

For a Gaussian optical beam with waist w_0 at frequency $f = c/\lambda$, the relevant quantities are:

- beam radius at distance z: $w_z = z \tan\left(\frac{c}{\pi f w_0}\right)$;
- peak collected power fraction: $A_0 = \frac{2w_d^2}{w_z^2}$;

- distance-dependent pointing jitter variance: $\sigma_s^2(z) = \sigma_{s0}^2 \exp\left(\frac{2k_0 z}{d_0}\right)$.

The pointing shape parameter α is:

$$\alpha = \frac{w_z^2}{4\sigma_s^2} \tag{3.17}$$

The radial pointing deviation is Rayleigh distributed. The pointing loss fraction $y \in (0, A_0]$ then follows a power-law probability density function (PDF) and cumulative distribution function (CDF):

$$f_Y(y) = \frac{\alpha}{A_0^\alpha} y^{\alpha-1}, \qquad F_Y(y) = \left(\frac{y}{A_0}\right)^\alpha, \qquad 0 < y \le A_0 \tag{3.18}$$

3.1.5 Achievable Data Rate Integration

The hard-decision information rate per symbol $m_i(g)$ [bits/symbol] as a function of SNR $g = E_s/N_0$ is formulated using the binary entropy:

$$H_2(p) = -p\log_2(p) - (1-p)\log_2(1-p) \tag{3.19}$$

- **BPSK (1 bit/symbol):** $m_{BPSK}(g) = 1 \cdot \left[1 - H_2\left(Q(\sqrt{2g})\right)\right]$
- **OOK (1 bit/symbol):** $m_{OOK}(g) = 1 \cdot \left[1 - H_2\left(Q(\sqrt{g})\right)\right]$. OOK is BPSK at $g/2$, because half of its energy is a data-free tone.
- **QPSK (2 bits/symbol):** $m_{QPSK}(g) = 2 \cdot \left[1 - H_2\left(Q(\sqrt{g})\right)\right]$

where $Q(x) = \frac{1}{2}\mathrm{erfc}\left(x/\sqrt{2}\right)$.

The soft-decision rates are evaluated by Gauss–Hermite quadrature. For BPSK,

$$m_{BPSK}^s(g) = 1 - \mathbb{E}_z \log_2\left(1 + e^{-4g(1+z/\sqrt{2g})}\right), \qquad z \sim \mathcal{N}(0,1),$$

and $m_{OOK}^s(g) = m_{BPSK}^s(g/2)$, $m_{QPSK}^s(g) = 2m_{BPSK}^s(g/2)$. These are upper bounds on the hard-decision rates.

Integrating over the pointing fading distribution gives the ergodic achievable rate R_{KK} [bits/s]:

$$R_{KK} = \int_0^{A_0} B \cdot m_i\big(g(y)\big)\, f_Y(y)\, dy \tag{3.20}$$

No detector threshold is imposed; the rate falls off through $g(y)$ itself. If a tracking threshold power p_{th} is to be modelled, the lower limit becomes $y_{th} = p_{th}/(h_{PL}P_T)$, which is dimensionless. Applying the logarithmic change of variable $t = \ln y$ ($y = e^t$, $dy = e^t dt$) yields the exact computational form:

$$R_{KK} = B \int_{-\infty}^{\ln A_0} m_i\big(g(e^t)\big) \cdot \alpha \exp\big(\alpha(t - \ln A_0)\big)\, dt \tag{3.21}$$

Pointing jitter is slow compared with a codeword, so the rate guaranteed for all but a fraction ε of the time is also reported. It follows in closed form from F_Y:

$$R_\varepsilon = B \cdot m_i\big(g(A_0\, \varepsilon^{1/\alpha})\big) \tag{3.22}$$

The receiver sensitivity is the received power at which the BER reaches the 7 % hard-decision forward error correction (FEC) threshold of 3.8×10^{-3}. This requires $g^\star = 5.5$ dB for BPSK and 8.5 dB for QPSK and OOK.

Table 3.1: DEFAULT PARAMETERS SETUP II

Parameter	**Value**	**Description**
w_0	0.1 m	beam waist at TX
w_d	0.07 m	detector radius (30 % smaller than w_0)
$f_{default}$	195.94 THz (1530 nm)	default frequency (ITU-T C-band short edge)
h_{PL}	0.9	deterministic path loss
η	0.5 A/W	detector responsivity
$P_{T,default}$	0.3 W	total transmit power (carrier + signal)
κ	12 dB	CSPR at transmitter
B	10 GHz	symbol rate / bandwidth (RRC, $\beta = 0.05$)
i_{th}	20 pA/$\sqrt{\text{Hz}}$	thermal noise current density
RIN	-150 dB/Hz	transmit-laser RIN
G, F, B_f	15 dB, 5 dB, 30 GHz	pre-amplifier gain, noise figure, optical filter
z	1000 km	link distance
$\sigma_{S0}; k_0; d_0$	2; 0.1; 100e3	σ_s model, paper given values

Table 3.2: Noise Budget at $y = A_0$ ($A_0 = 4.132 \times 10^{-4}$, $w_z = 4.870$ m, $\alpha = 0.201$)

Received power (carrier / signal)	-9.52 dBm (-9.79 / -21.79 dBm)
KK noise factor ζ	1.126
SNR (shot only)	47.4 dB
SNR (thermal only)	48.9 dB
SNR (RIN only)	40.0 dB
SNR (signal/carrier–ASE only)	31.9 dB
SNR (ASE×ASE only)	74.8 dB
KK reconstruction (QPSK, waveform, noiseless)	100.9 dB (negligible)
SNR (total, on-axis, pre-amp)	31.1 dB
SNR (total, photodiode only, no pre-amp)	18.6 dB
Sensitivity BPSK / QPSK (pre-amp)	-30.4 / -28.6 dBm

3.2 Simulation and Results

Worked check of Table 3.2. At $y = A_0$ the received power is $h_{PL}P_T A_0 = 0.9 \times 0.3 \times 4.132 \times 10^{-4} = 1.116 \times 10^{-4}$ W, i.e. -9.52 dBm. Of this, $\kappa/(1+\kappa) = 0.941$ is carrier. With the photodiode alone:

- cross-beat signal power: $S = 2(0.5)^2(1.050 \times 10^{-4})(6.624 \times 10^{-6}) = 3.478 \times 10^{-10}$ A^2;
- thermal noise power: $\zeta i_{th}^2 B = 1.126 \times 4 \times 10^{-22} \times 10^{10} = 4.50 \times 10^{-12}$ A^2;
- ratio: 77.2, i.e. 18.9 dB.

With the 15 dB pre-amplifier the thermal SNR rises by 2×15 dB to 48.9 dB, and the receiver becomes ASE limited (31.9 dB).

Ergodic rates. At 195.94 THz (1530 nm) and $P_T = 0.3$ W, with the pre-amplifier, the ergodic rates are:

- BPSK: 6.84 Gbps (hard decision) / 6.96 Gbps (soft decision);

- OOK: 6.59 / 6.72 Gbps;
- QPSK: 13.18 / 13.44 Gbps.

Without the pre-amplifier they fall to 3.80 (BPSK) and 6.70 Gbps (QPSK, hard decision).

Monte Carlo validation. Averaging $Bm(g(y))$ over the 2×10^5 channel samples gives 13.19 Gbps for QPSK, matching the integral (13.18 Gbps). The end-to-end waveform simulation (RRC pulses, carrier plus SSB signal, ASE in both polarisations, square-law photodiode with shot, thermal and RIN noise, KK DSP, matched filter and a 512-symbol pilot estimate) agrees with the analytic $g(y)$ within 0.1 dB from $y = A_0$ to $y = 0.03A_0$; at $y = 0.01A_0$ (SNR ≈ 7 dB) the waveform is 0.4 dB lower, because the KK logarithm amplifies noise when the photocurrent approaches zero. With 2^{13} symbols, bit errors are observed only at $y = 0.01A_0$ (BER 1.20×10^{-2} measured vs 1.22×10^{-2} analytic); at the other points the validation rests on the measured SNR.

For the photodiode-only receiver at low SNR, the photocurrent noise approaches the carrier level. The KK logarithm then amplifies it, and the analytic rate becomes an upper bound. This can be checked with the waveform Monte Carlo by setting `pre_on=false`.

CSPR optimisation. Fig. 3.6 shows the waveform SNR as a function of κ for three pointing states, together with the analytic model. The optimum is $\kappa^\star \approx 7$ dB at $y = A_0$ and $0.1A_0$ (34.2 and 23.3 dB, against 31.1 and 20.9 dB at 12 dB) and $\kappa^\star \approx 9$ dB at $y = 0.03A_0$ (16.2 vs 14.3 dB). Operating at the optimum gains 2–3 dB; the optimum moves to higher κ in deep fades. Below the optimum the analytic model, which does not capture KK breakdown, overestimates the SNR. The measured optimum of Wen *et al.* [6], 7–8 dB, agrees. This report keeps $\kappa = 12$ dB, where the noise model is verified.

Outage. The ε-outage rate is effectively zero for $\varepsilon = 0.1$ ($R_{0.1} \approx 7.6\times10^{-5}$ Gbps). Under the jitter model, $\sigma_s = 5.44$ m exceeds the beam radius $w_z = 4.87$ m at 1000 km, so $\alpha = 0.20$. With the smaller detector radius $w_d = 0.07$ m, $A_0 = 4.13\times10^{-4}$ and the link spends 41 % of the time below the QPSK sensitivity even with the pre-amplifier. Pointing jitter, not the KK receiver, limits reliability.

3.2.1 Constellation Analysis: Carrier Placement Configurations

To demonstrate *why* the carrier must co-propagate with the data, four carrier/LO configurations are simulated at $y = A_0$ for QPSK with 2^{13} symbols. All four use the same waveform chain; the gain and phase are estimated from 512 pilot symbols, and SNR, EVM, BER and phase error are measured on the remaining 7680 symbols.

- **A2 (TX Carrier):** The standard configuration used throughout this chapter. The carrier co-propagates with the signal through the channel, undergoes the same fading and phase rotation, and arrives at the photodiode in phase with the data.
- **B1 (RX Optical LO):** The transmitter sends only the SSB data signal at full power P_T. A local CW laser of 1.77 mW (CSPR 12 dB at $y = A_0$) is combined with the incoming signal *before* the photodiode; no EDFA is used, since the LO provides the gain. The LO is ideal: no frequency offset, no linewidth and matched polarisation.

- **A1/B3 (No Carrier):** The signal is transmitted and received alone, with no carrier from either end. The KK algorithm attempts phase retrieval from an intensity that does not satisfy the minimum-phase condition.
- **B2 (Carrier added after detection):** The signal is received without a carrier, and the carrier level is added digitally to the photocurrent before the KK algorithm. This tests whether the carrier can be supplied in DSP instead of optically.

Table 3.3: Constellation study at $y = A_0$, QPSK (7680 payload symbols, 15 360 bits)

Configuration	**SNR**	**EVM**	**BER**	σ_ϕ
A2 TX carrier	31.1 dB	2.8 %	0/15 360	0.020 rad
B1 RX LO (ideal)	37.3 dB	1.4 %	0/15 360	0.009 rad
A1/B3 no carrier	−27.9 dB	> 100 %	0.51	1.82 rad
B2 carrier added after detection	−24.4 dB	> 100 %	0.51	1.83 rad

Configurations A2 and B1 both produce clean QPSK constellations, confirming that KK phase retrieval works whenever a carrier is present at the photodiode input, whether it arrived through the channel (A2) or was added locally (B1). A2 agrees with the analytic SNR (31.05 dB). B1 is 6.2 dB better because all of the transmit power carries data and its LO is ideal; this is the CSPR penalty of the transmitter carrier, measured against a receiver LO that has no frequency offset, no phase noise and no polarisation mismatch. With a realistic LO (100 MHz residual offset after locking, 200 kHz combined linewidth) and the same frequency and phase recovery as A2, B1 is still 4.2 dB better on-axis and 6.9–8.7 dB better at $y \leq 0.1A_0$, because its fixed LO power keeps the beat signal above thermal noise while the transmitted carrier fades with the data. The case for the transmitter carrier therefore rests on the hardware it avoids, not on SNR. Configurations A1/B3 and B2 fail completely: the BER is 0.5 and the phase error is spread over $[-\pi, \pi]$. Square-law detection $I = \eta|E|^2$ destroys the optical phase; the Hilbert-transform relation (eq. 3.7) recovers it only if a dominant carrier is present in the optical field *before* detection. Adding the carrier level digitally afterwards (B2) cannot restore information that the photocurrent no longer contains.

3.2.2 Detector Radius Sensitivity

The detector radius w_d appears in the peak collected fraction $A_0 = 2w_d^2/w_z^2$ and therefore directly controls the on-axis received power. Following the supervisor's guidance, w_d is set 20–50 % below the transmitter beam waist $w_0 = 0.1$ m, which motivates a sweep from 0.04 m to 0.10 m. The ergodic QPSK rate rises from 11.46 Gbps at $w_d = 0.04$ m to 14.09 Gbps at $w_d = 0.10$ m; the default $w_d = 0.07$ m gives 13.18 Gbps, about 94 % of the rate at the largest radius in the sweep. The rate keeps increasing with w_d.

3.2.3 ITU-T C-Band Wavelength Comparison

The ITU-T C-band spans 1530–1565 nm (191.56–195.94 THz). At the long-wavelength edge (1565 nm) the beam divergence is slightly larger ($w_z = 4.98$ m vs 4.87 m), which reduces A_0 from 4.13×10^{-4} to 3.95×10^{-4}. However, the pointing shape parameter α increases from 0.201 to 0.210, which means the fading distribution is slightly less severe. These two effects approximately cancel: the ergodic QPSK rate changes from 13.18 Gbps (1530 nm) to 13.45 Gbps (1565 nm), a difference of about 2 % (2.3 % at $P_T = 0.1$ W).

3.3 Rationale: Transmitter-Side Carrier vs Receiver-Side Local Oscillator

Kramers–Kronig receivers were originally proposed for fibre-optic links, where the CW carrier can be placed either at the transmitter (co-propagating with the data) or at the receiver (as a local oscillator, LO). Both configurations recover the full optical field from a single photodiode, but they differ sharply in what the receiver must provide. This section compares the two options for the inter-satellite link scenario and motivates the choice of a transmitter-side carrier used throughout this chapter.

3.3.1 Receiver-Side LO

In the receiver-LO configuration the transmitter sends only the SSB data signal at full power P_T. The receiver mixes the incoming signal with a locally generated CW laser before the photodiode. The LO power can be chosen freely and made large, so the cross-beat current is proportional to $\sqrt{P_{LO}P_s}$ with an arbitrarily strong P_{LO}. This gives the architecture several advantages:

- **No CSPR penalty.** The full transmit power carries data; there is no power budget split between carrier and signal.
- **Shot-noise-limited sensitivity.** A strong LO lifts the beat current well above thermal noise, approaching coherent-receiver performance from a single photodiode.
- **No minimum-phase constraint.** The carrier power is set at the receiver and can always be made large enough; the PAPR of the transmitted signal is irrelevant to phase reconstruction.

However, in a free-space inter-satellite link these advantages come at significant cost:

- **Frequency locking.** The LO laser must be offset-locked to the transmitter laser to within a fraction of B. Two independent lasers on different spacecraft, each subject to mechanical vibration and thermal drift, require either an optical phase-locked loop (OPLL) or a digital frequency-offset estimator with $\geq$10-bit ADC and real-time DSP.
- **Doppler compensation.** For a 550 km LEO constellation the carrier Doppler shift can reach ± 9.9 GHz. The LO must track this shift in real time, which demands a widely tunable laser or an acousto-optic frequency shifter, adding mass, power and complexity.
- **Phase noise.** The transmitter and receiver lasers have independent phase-noise processes. The beat signal carries their sum, which must be tracked by carrier-phase recovery DSP. For linewidths above $\sim$ 100 kHz the cycle-slip rate becomes significant at the symbol rates considered here.
- **Polarisation.** The received field arrives in an arbitrary and time-varying polarisation state (due to spacecraft rotation and optical-bench flexure). A single photodiode with LO mixing sees only the projection onto the LO's polarisation; recovering the full signal requires a polarisation-diversity receiver (two photodiodes, a polarisation beam splitter and Stokes-parameter tracking), roughly doubling the front-end hardware.

3.3.2 Transmitter-Side Carrier

In the transmitter-carrier configuration the CW tone is generated from the *same laser* as the data signal and is transmitted in the same spatial mode, same polarisation and same beam. At the receiver, a single photodiode performs square-law detection; the KK algorithm recovers the optical field entirely in digital signal processing.

Because the carrier and data share the same laser and propagation path:

- **Phase noise cancels.** The laser phase $\theta_L(t)$ is common to the carrier and the signal. In the photocurrent $I(t) = \eta |E_{Rx}|^2$, the phase drops out. No carrier-phase recovery is needed.
- **Doppler cancels.** The Doppler shift $\phi_D(t)$ affects the carrier and the signal identically. The cross-beat current sees only the difference, which is the fixed SSB offset f_{sh}. No Doppler tracking or tunable laser is required.
- **Polarisation cancels.** Whatever polarisation rotation the channel imposes acts on the carrier and signal equally. The photocurrent is insensitive to the absolute polarisation state. A single photodiode suffices.
- **No LO laser.** The receiver is simply a photodiode (optionally preceded by an EDFA) followed by an ADC and DSP. There is no second laser, no frequency-locking loop, no polarisation optics.

The price is the CSPR power penalty. At $\kappa = 12\,\text{dB}$, only $P_s/P_T = 1/(1+\kappa) \approx 6\,\%$ of the transmit power carries information; the rest is spent on the carrier. Additionally, the minimum-phase condition $\kappa > \text{PAPR}_s$ imposes a floor on the CSPR, limiting the amount of power that can be shifted from carrier to data.

3.3.3 Trade-off Summary

Table 3.4: Comparison of transmitter-side carrier and receiver-side LO for OISL

Attribute	**TX-side carrier**	**RX-side LO**
Frequency locking	Not needed	Required (OPLL or DSP)
Doppler tracking	Not needed	$\pm 9.9\,\text{GHz}$ tuning for LEO
Phase-noise tolerance	Immune (common path)	Limited by combined linewidth
Polarisation control	Not needed	Diversity receiver or tracking
Receiver hardware	Photodiode + EDFA + ADC	Photodiode + tunable laser + PBS + 2 ADCs
Transmit power efficiency	$1/(1+\kappa) \approx 6\,\%$ at 12 dB	100 % (no carrier overhead)
Sensitivity (pre-amp)	ASE limited ($-28.6\,\text{dBm}$ QPSK)	Near shot-noise limited
Min-phase constraint	$\kappa > \text{PAPR}_s$ required	None
DSP complexity	KK: ln, Hilbert, matched filter	Freq. offset + phase recovery + pol. demux

For a mass- and power-constrained satellite terminal, the transmitter-side carrier trades about 6 dB of SNR (6.2 dB against an ideal receiver LO in Section 3.2.1) for the complete

elimination of the LO laser, frequency-locking loop, polarisation optics and Doppler-tracking hardware. The resulting receiver is a single photodiode with digital back-end, which is the simplest coherent-capable architecture possible. The CSPR power penalty is partially recovered by the pre-amplifier (which boosts carrier and signal equally) and could be reduced further by operating near the optimal CSPR found by waveform simulation (7–9 dB, a gain of 2–3 dB over 12 dB). In constellations where the link budget has margin (e.g. short LEO–LEO links with < 2000 km range), the power penalty is acceptable and the hardware simplification is decisive.

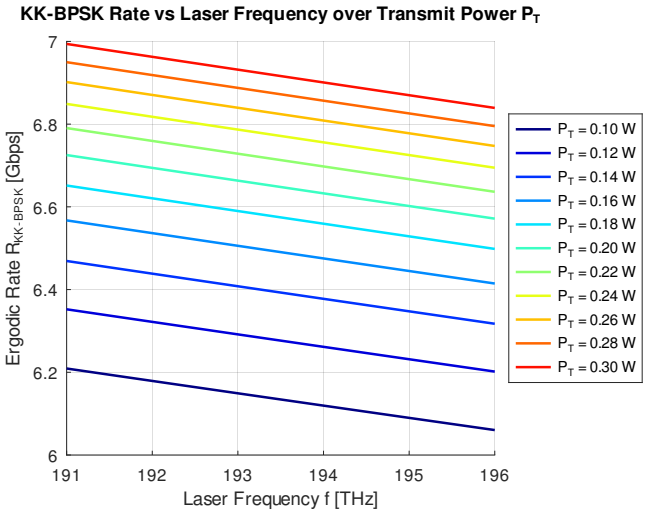


Figure 3.1: BPSK Data Rate vs frequency

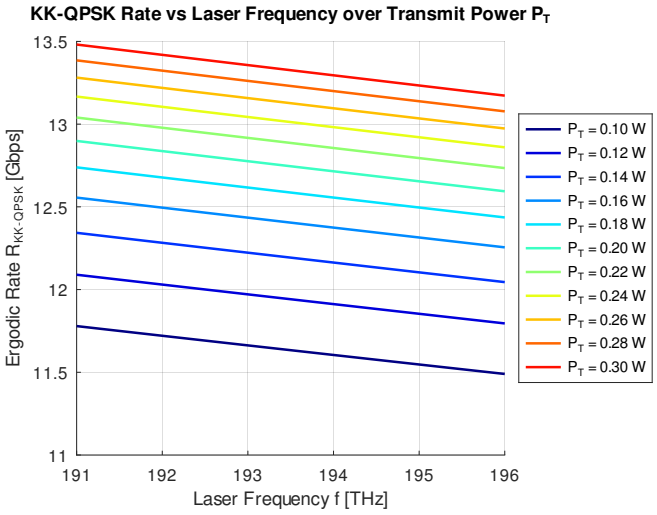


Figure 3.2: QPSK Data Rate vs frequency

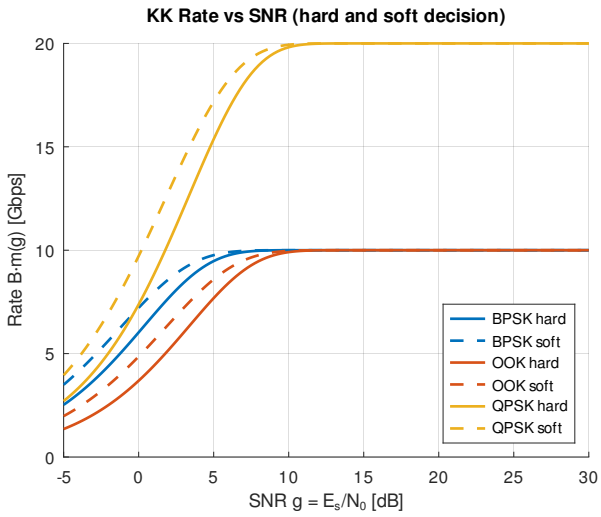


Figure 3.3: Data Rate vs SNR (solid: hard decision, dashed: soft decision)

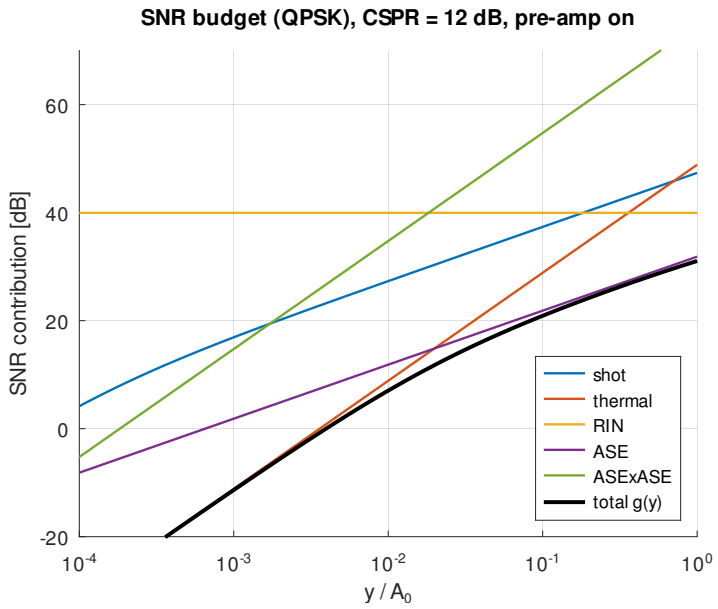
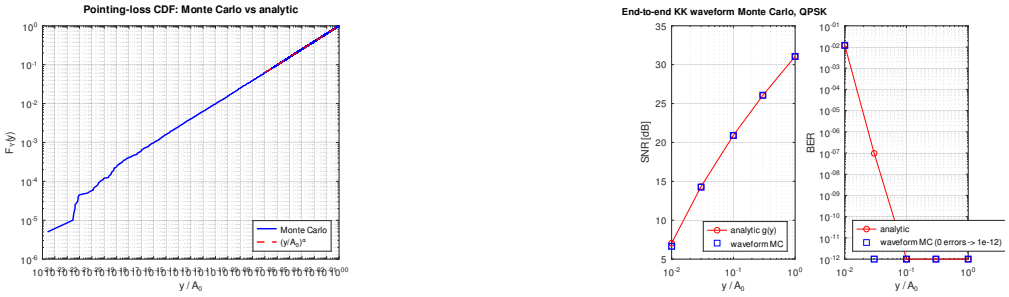


Figure 3.4: SNR contributions vs pointing fraction y (QPSK, $\kappa = 12\,\mathrm{dB}$)



(a) (b)

Figure 3.5: Monte Carlo validation: (a) pointing-loss CDF from 2×10^5 Rayleigh samples; (b) end-to-end KK waveform simulation (2^{13} symbols per point, 512 pilots) vs analytic $g(y)$, QPSK

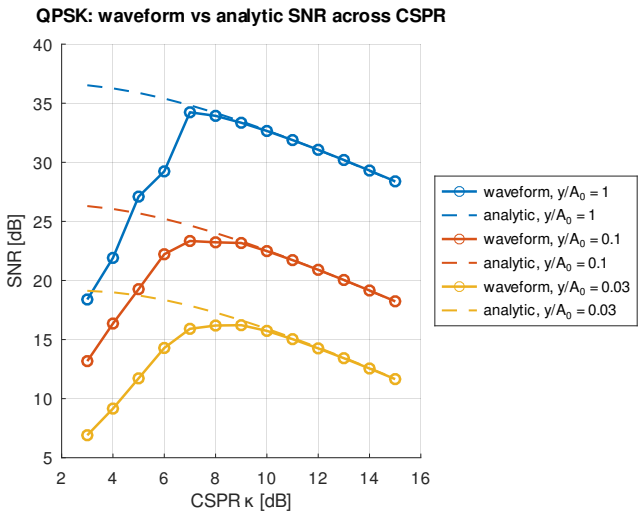


Figure 3.6: QPSK SNR vs CSPR: waveform simulation (solid) and analytic model (dashed) at $y/A_0 = 1$, 0.1 and 0.03.

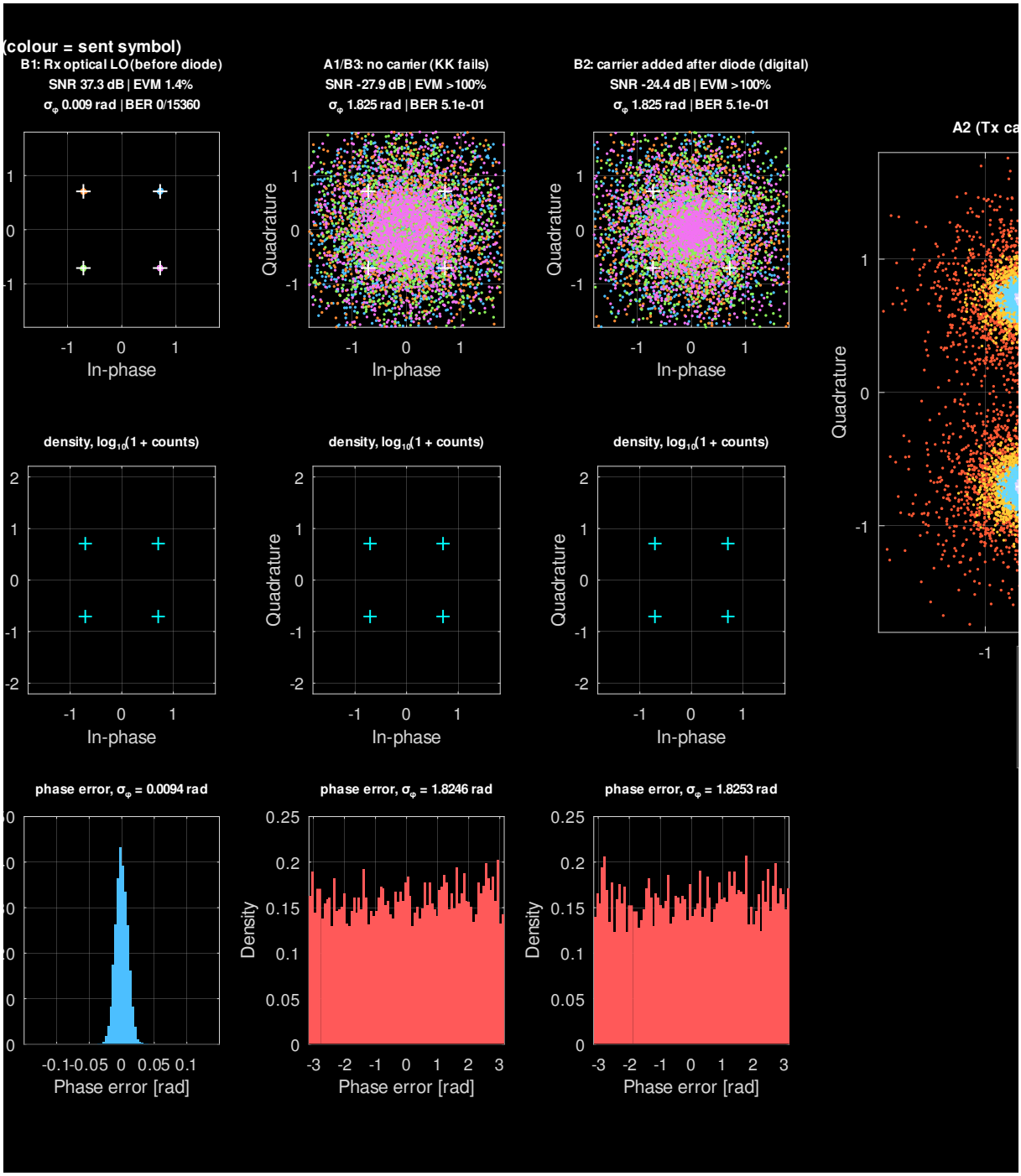


Figure 3.7: QPSK constellation study at $y = A_0$. Columns 1–4: A2, B1, A1/B3 and B2, each with the constellation (colour = transmitted symbol), a density map and the phase-error distribution. Right: A2 at $y/A_0 = 1$, 0.1, 0.03 and 0.01.

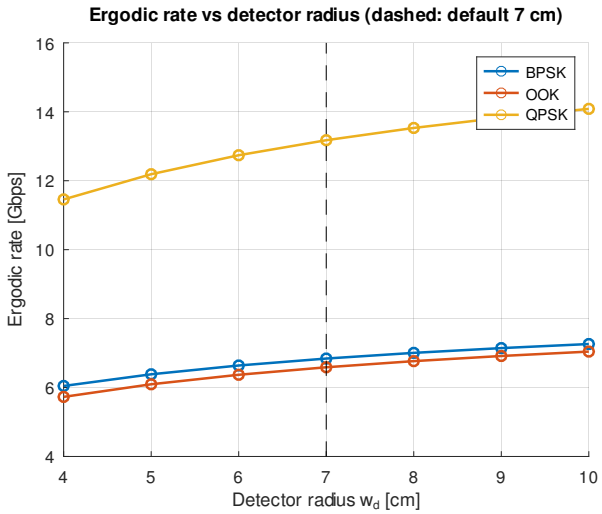


Figure 3.8: Ergodic rate vs detector radius w_d for all three modulation formats. Vertical dashed line: default $w_d = 0.07\,\text{m}$. The rate increases monotonically with w_d but with diminishing returns above $w_d \approx 0.08\,\text{m}$.

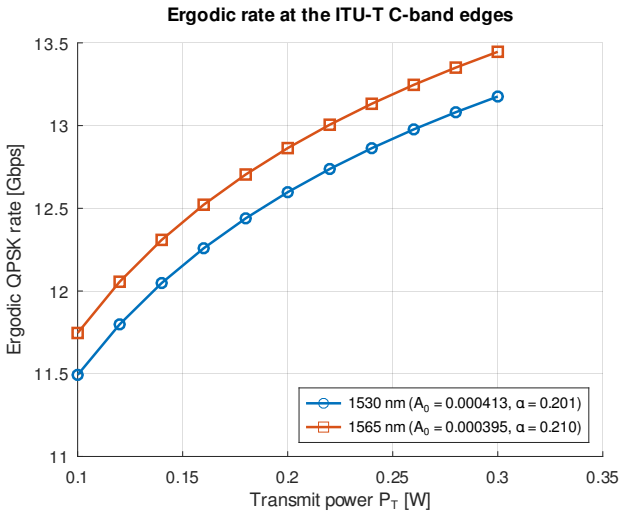


Figure 3.9: Ergodic rate vs transmit power P_T at the two ITU-T C-band edges: 1530 nm and 1565 nm. The 1565 nm curve is slightly higher due to the more favourable α despite lower A_0. The difference is about 2 % across the power range.

Chapter 4

Conclusion and Future Work

4.1 Summary of Findings

This report presented an end-to-end analytical model for an optical inter-satellite link (OISL) using a Kramers–Kronig (KK) self-coherent receiver with a transmitter-side carrier. The key findings are:

1. **Achievable rates.** At a link distance of 1000 km, transmit power $P_T = 0.3$ W and ITU-T C-band wavelength 1530 nm, the ergodic QPSK rate is 13.18 Gbps (hard decision) and 13.44 Gbps (soft decision) with an EDFA pre-amplifier. Without the pre-amplifier the rate drops to 6.70 Gbps, because the receiver becomes thermal-noise limited rather than ASE limited.

2. **Noise model and CSPR optimisation.** The analytic noise model agrees with the end-to-end waveform simulation within 0.15 dB for $\kappa \geq 12$ dB. A closed-form Gaussian bound on the KK reconstruction penalty, $\mathrm{PAPR}_s \cdot P_{viol}(\kappa)$, turned out 35–50 dB too pessimistic and is not used. The waveform simulation places the optimum CSPR at about 7 dB on-axis and 9 dB in deep fades, 2–3 dB better than the 12 dB design value, in agreement with the 7–8 dB measured by Wen *et al.*

3. **Carrier placement.** The constellation-level comparison of four carrier/LO configurations confirmed that KK phase retrieval requires a carrier at the photodiode input. Configurations without an optical carrier at the photodiode (A1/B3, and B2 where the carrier is added digitally after detection) fail because square-law detection destroys optical phase irreversibly; the Hilbert-transform relation holds only when the field is minimum-phase. The transmitter-side carrier (A2) and receiver-side optical LO (B1) both produce clean constellations. An ideal receiver LO is 6.2 dB better, which is the CSPR penalty; only the transmitter-side carrier cancels Doppler, phase noise and polarisation rotation without additional hardware.

4. **Detector radius.** The ergodic QPSK rate increases from 11.46 Gbps at $w_d = 0.04$ m to 14.09 Gbps at $w_d = 0.10$ m. The default $w_d = 0.07$ m (30 % smaller than the transmitter waist w_0) reaches 94 % of the rate at the largest radius in the sweep.

5. **ITU-T C-band.** The rate varies by about 2 % between 1530 and 1565 nm. Slightly higher beam divergence at 1565 nm is compensated by a more favourable pointing shape parameter α, so the system operates essentially flat across the C-band.

6. **Reliability.** Under the jitter model of Shang *et al.*, the pointing jitter $\sigma_s = 5.44\,\mathrm{m}$ exceeds the beam radius $w_z = 4.87\,\mathrm{m}$ at 1000 km, yielding $\alpha = 0.20$. The link spends 41 % of the time below the QPSK sensitivity even with the pre-amplifier. Pointing jitter, not the KK receiver architecture, is the reliability bottleneck.

7. **Monte Carlo validation.** Channel-level Monte Carlo (200 000 samples) matched the ergodic rate integral to within 0.02 Gbps. The waveform simulation (full KK DSP with pulse shaping, ASE, shot, thermal and RIN noise) agreed with the analytic SNR within 0.1 dB from $y = A_0$ to $0.03A_0$ and within 0.4 dB at $0.01A_0$.

4.2 Limitations

The model makes several simplifying assumptions:

- The pointing error is two-axis Rayleigh with a parametric jitter model. Measured pointing statistics from operational spacecraft may differ.
- The EDFA gain is treated as fixed and noiseless apart from ASE. Gain saturation and transient effects under fading are not modelled.
- Atmospheric effects are absent (vacuum propagation). For links involving a ground station, turbulence-induced scintillation and beam wander would need to be added.
- Only single-carrier modulation is considered. Wavelength-division multiplexing (WDM) with multiple KK channels could increase aggregate throughput.
- The analytic noise model is verified for $\kappa \geq 12\,\mathrm{dB}$ only; below that, results come from the waveform simulation. No validated closed-form model of the KK reconstruction penalty is available.
- The default jitter ($\sigma_{s0} = 2\,\mathrm{m}$, $\alpha = 0.20$) is a stress case taken from Shang *et al.*; realistic pointing for operational terminals has not yet been evaluated.
- The residual Doppler offset and clock scaling are simulated with pilot-based recovery, but oscillator mismatch between transmitter and receiver (which requires timing recovery in any receiver) and Doppler rate within a frame are not.
- The receiver-LO comparison models LO frequency offset and linewidth but not polarisation mismatch, and the LO phase tracker is tuned per case (optimistic for B1).
- The jitter model gives σ_s as a length; with angular jitter, α would not depend on range.
- With 2^{13} symbols per waveform run, bit-error rates below about 10^{-4} cannot be measured directly; dedicated runs with about 10^6 symbols confirm the analytic BER at 3.8×10^{-3} and 10^{-4}.
- Hardware effects (modulator and driver nonlinearity, photodiode/TIA bandwidth, ADC resolution, EDFA gain dynamics) are not modelled.

4.3 Future Work

Several extensions would strengthen the model and bring it closer to a system-level design tool:

1. **OptiSystem co-simulation.** The self-contained MATLAB function `oisl_kk_channel.m` is ready for integration with OptiSystem's MATLAB co-simulation interface. A logical next step is to run the waveform-level simulation inside OptiSystem's signal-processing framework, connecting the analytical pointing-error model to OptiSystem's FSO channel module.

2. **Higher-order modulation.** The model supports BPSK, OOK and QPSK. Extending to 16-QAM would test the limits of the KK approach at higher spectral efficiencies, where the reconstruction penalty grows steeper.

3. **Adaptive CSPR.** The optimal CSPR depends on the link budget (and therefore on range and pointing state). An adaptive scheme that adjusts the CSPR at the transmitter based on a feedback channel could improve the average throughput.

4. **Multi-link scenarios.** In a constellation with relay hops, the end-to-end rate depends on the weakest link. Extending the model to a multi-hop OISL network with heterogeneous ranges and pointing conditions would be valuable for constellation design.

5. **Realistic pointing.** With $\sigma_{s0} = 0.5$ m, 1 % outage at 1000 km needs only 11.9 dBm (13 dB margin at 0.3 W) and QPSK saturates at 20 Gb/s; receiver sensitivity, and so the CSPR penalty, then sets the power budget. A study under an angular jitter model, and a comparison with coherent and IM/DD receivers under the same fading, would show where the transmitter carrier is the right choice.

6. **Front-end distortion and MP-KK.** Once OptiSystem component models add hardware nonlinearity and memory, compare the conventional KK receiver with GKK and MP-KK pre-compensation [6]; the simulation code for this comparison is already in place.

7. **Experimental validation.** A benchtop demonstration of the transmitter-side-carrier KK receiver at 10 GBaud, even without the free-space channel, would validate the noise model and the CSPR optimum against measured data.

Bibliography

[1] A. Mecozzi, C. Antonelli, and M. Shtaif, "Kramers–Kronig coherent receiver," *Optica*, vol. 3, no. 11, pp. 1220–1227, 2016.

[2] B. Shang, S. Zhang, and Z. J. Wong, "Channel modeling and rate analysis of optical inter-satellite link (OISL)," *IEEE Trans. Veh. Technol.*, vol. 74, no. 7, pp. 11650–11655, 2025, doi: 10.1109/TVT.2025.3549724.

[3] H. Kaushal and G. Kaddoum, "Optical communication in space: Challenges and mitigation techniques," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 1, pp. 57–96, 2017.

[4] A. Mecozzi, C. Antonelli, and M. Shtaif, "Kramers–Kronig receivers," *Adv. Opt. Photon.*, vol. 11, no. 3, pp. 480–517, 2019.

[5] M. Naghshvarianjahromi, S. Kumar, and M. J. Deen, "Free space ground to satellite optical communications using Kramers–Kronig transceiver in the presence of atmospheric turbulence," *Sensors*, vol. 22, no. 9, 3435, 2022, doi: 10.3390/s22093435.

[6] Z.-Y. Wen, F. Yang, H. Jiang, Y. Liu, F. Chen, M. Li, and J. Zhang, "Modified Kramers–Kronig receiver based on memory polynomial compensation for photonics-assisted millimeter-wave communications," *Opt. Express*, vol. 31, no. 21, pp. 34800–34816, 2023, doi: 10.1364/OE.501828.